

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the October/November 2006 question paper

9702 PHYSICS

9702/04

Paper 4 (Core), maximum raw mark 60

This mark scheme is published as an aid to teachers and students, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

The grade thresholds for various grades are published in the report on the examination for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses.

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	GCE A/AS LEVEL - OCT/NOV 2006	9702	04

- 1 (a) *either* ratio of work done to mass/charge
or work done moving unit mass/charge from infinity
or both have zero potential at infinity B1 [1]
- (b) gravitational forces are (always attractive) B1
electric forces can be attractive or repulsive B1
for gravitational, work got out as masses come together
/mass moves from infinity B1
for electric, work done on charges if same sign, work got out if opposite sign as charges
come together B1 [4]
- 2 (a) (i) idea of heat lost (by oil) = heat gained (by thermometer) C1
 $32 \times 1.4 \times (54 - t) = 12 \times 0.18 \times (t - 19)$ C1
 $t = 52.4^\circ\text{C}$ A1 [3]
- (ii) *either* ratio $(= 1.6/54) = 0.030$ or $(= 1.6/327) = 0.0049$ A1 [1]
- (b) thermistor thermometer (allow 'resistance thermometer') B1
because small mass/thermal capacity B1 [2]
- (c) boiling point temperature is constant M1
further comment
e.g. heating of bulb would affect only rate of boiling A1 [2]
- 3 (a) use of $a = -\omega^2 x$ clear C1
either $\omega = \sqrt{(2k/m)}$ or $\omega^2 = (2k/m)$ B1
 $\omega = 2\pi f$ C1
 $f = (1/2\pi)\sqrt{(2 \times 300)/0.240}$ B1
 $= 7.96 \approx 8 \text{ Hz}$ A0 [4]
- (b) (i) resonance B1 [1]
- (ii) 8 Hz B1 [1]
- (c) (increase amount of) damping B1
without altering (k or) m ... (some indirect reference is acceptable) B1
sensible suggestion B1 [3]
- 4 (a) (i) $\frac{GMm}{1/2m} \{(R + h_1)^{-1} - (R + h_2)^{-1}\}$ B1
 $\{v_1^2 - v_2^2\}$ B1 [2]
- (b) $2M \times 6.67 \times 10^{-11} \{(26.28 \times 10^6)^{-1} - (29.08 \times 10^6)^{-1}\} = 5370^2 - 5090^2$ B1
 $M \times 4.888 \times 10^{-19} = 2.929 \times 10^6$ C1
 $M = 6.00 \times 10^{24} \text{ kg}$ A1 [3]
(If equation in (a) is dimensionally unsound, then 0/3 marks in (b), if dimensionally sound but incorrect, treat as e.c.f.)
- 5 (a) (i) (induced) e.m.f proportional/equal to rate of change of flux (linkage) B1
(allow 'induced voltage, induced p.d.')
flux is cut as the disc moves M1
hence inducing an e.m.f A0 [2]
- (ii) field in disc is not uniform/rate of cutting not same/speed of disc not same (over whole disc) B1
so different e.m.f.'s in different parts of disc M1
lead to eddy currents A0 [2]
- (b) eddy currents dissipate thermal energy in disc B1
energy derived from oscillation of disc B1
energy of disc depends on amplitude of oscillations B1 [3]

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- 6 (a) (i) peak voltage = $6\sqrt{2}$
peak voltage = 8.48 V C1 A1 [2]
- (ii) zero because *either* no current in circuit (and $V = IR$)
or all p.d. across diode B1 [1]
- (b) waveform: half-wave rectification B1
peak height at about 4.25 cm B1
half-period spacing of 2.0 cm B1 [3]
(allow $\pm\frac{1}{4}$ square for height and half-period)
- (c) (i) capacitor shown in parallel with resistor B1 [1]
- (ii) *either* energy = $\frac{1}{2}CV^2$ or = $\frac{1}{2}QV$ and $Q = CV$ C1
 $= \frac{1}{2} \times 180 \times 10^{-6} \times (6\sqrt{2})^2$ C1
 $= 6.48 \times 10^{-3} \text{ J}$ A1 [3]
- (iii) *either* fraction = 0.43^2 or final energy = 1.2 mJ C1
fraction = 0.18 A1 [2]
- 7 (a) (i) quantum/packet/discrete amount of energy M1
electromagnetic mentioned A1 [2]
- (ii) max. k.e. corresponds to electron emitted from surface B1
energy is required to bring electron to surface B1 [2]
- (b) at higher frequency, fewer photons (per second) for same intensity M1
so rate of emission decreases A1 [2]
(allow argument based on photoelectric efficiency)
- 8 (a) (i) *either* number = $6.02 \times 10^{23} \times \{2.65 \times 10^{-6}\}/234$ C1
or number = $(2.65 \times 10^{-9})/(234 \times 1.66 \times 10^{-27})$ A1 [2]
 $= 6.82 \times 10^{15}$
- (ii) $A = \lambda N$ C1
 $604 = \lambda \times 6.82 \times 10^{15}$
 $\lambda = 8.86 \times 10^{-14} \text{ s}^{-1}$ A1 [2]
- (iii) $T_{1/2} = \ln 2 / \lambda$ C1
 $= 7.82 \times 10^{12} \text{ s}$ A1 [2]
 $= 2.48 \times 10^5 \text{ years}$
- (b) half-life is (very) long (compared with time of counting) B1 [1]
- (c) there would be appreciable decay of source during the taking of measurements B1 [1]